Product Requirement Document (PRD)
EduTrace■ES

1. Product Overview

EduTrace-ES is an Academic Accreditation Status Tracing Platform designed to help educational institutions monitor and improve their accreditation readiness.

The system integrates embedded systems, environmental sensors, academic databases, and artificial intelligence to analyze institutional performance and generate an accreditation readiness score.

EduTrace-ES helps institutions track parameters related to NAAC/NBA accreditation, monitor infrastructure conditions, analyze academic documentation, and provide AI-based recommendations for improvement.

2. Objectives

• Monitor institutional infrastructure and environmental parameters

• Analyze academic documents and reports

• Simulate accreditation readiness score

• Provide AI-based improvement suggestions

• Track progress toward accreditation standards

3. Target Users

• Educational Institutions (Colleges & Universities)

• Accreditation Committees

• Institution Administrators

• Quality Assurance Cells (IQAC)

4. Key Features

Accreditation Readiness Scoring:

The platform analyzes institutional data and generates a readiness score based on parameters such as innovation and research, SDG values, IIQA, SSR, DVV, student satisfaction survey, bibliometric data, and institutional performance indicators.

Embedded System Monitoring:

EduTrace-ES uses microcontroller-based embedded systems to monitor infrastructure and environmental conditions.

Sensors include:

• Current Sensor – monitors power usage and equipment activity

• Temperature Sensor – tracks environmental temperature

• Motion Sensor – monitors activity in labs or classrooms

• Camera Module – visual monitoring and verification

• Moisture Sensor – detects environmental conditions

• AQI Sensor (optional) – monitors air quality

AI Document Analysis:

Artificial Intelligence analyzes institutional documents such as accreditation reports, research publications, and institutional records to identify insights and documentation gaps.

Predictive Quality Analysis:

AI models evaluate institutional data to predict accreditation performance, quality benchmarks, and improvement areas.

Analytics Dashboard:

Displays accreditation readiness score, department performance, sensor insights, document analysis results, and institutional progress tracking.

AI Improvement Suggestions:

The system provides recommendations such as research improvements, documentation gaps, innovation opportunities, and environmental improvements.

5. System Components

Hardware:

• Microcontroller

• Environmental Sensors

• Camera Modules

• Network Communication Modules

Software:

• AI Document Analysis Engine

• Data Analytics Dashboard

• Institutional Data Integration System

• Accreditation Scoring Algorithm

6. Success Metrics

• Accuracy of accreditation readiness prediction

• Improvement in institutional accreditation scores

• Reduction in accreditation preparation time

• Number of institutions adopting the system

7. Expected Outcome

EduTrace-ES helps institutions continuously monitor and improve their accreditation readiness, reducing manual workload and enabling data■driven quality improvement.